MAAT v1.2.1 Structural Signature Test in Growth Data

Projection CCI, Residual Magnitudes, and Exploratory Null Tests

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Abstract

This paper performs a structural signature test of the MAAT v1.2.1 framework using a compact set of $f\sigma_8(z)$ growth measurements. The goal is not to improve a cosmological likelihood, but to ask whether MAAT projection diagnostics align with residual structure relative to a Planck-normalised ΛCDM growth baseline.

For each data point we compute signed residuals $r_\sigma = (f\sigma_{8,\mathrm{obs}} - f\sigma_{8,\Lambda\mathrm{CDM}})/\sigma_i$ and their magnitudes $|r_\sigma|$. We then compare these residuals with projection and closure diagnostics, including $\mathcal{C}_{\mathrm{proj}}$, R_{proj}, (H, B, S, V), R_{resp}, R_{rob}, $\mathrm{CCI}_{\mathrm{min}}$, and $\mathrm{CCI}_{\mathrm{diag}}$.

The most relevant diagnostic result is

$$\rho_{\mathrm{S}}(\mathrm{CCI}_{\mathrm{diag}}, |r_\sigma|) = 0.5934, \qquad p_{\mathrm{perm}} = 0.0338,$$

with comparable random-field and redshift-shuffle null tests $p \simeq 0.036$. This is an exploratory structural-signature result for $N = 13$ points, not a detection and not a full cosmological likelihood. Because the balance support B is constructed from residual information in this benchmark, the result should be interpreted as a semi-supervised diagnostic consistency test rather than a fully independent blind prediction.

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1 Purpose and Status

Papers 33–39 developed the projection observable, v1.2.1 robustness closure, and a small observable-signature proxy for $f\sigma_8$. Paper 40 asks a different question:

> *Do MAAT structural diagnostics track where the reference* ΛCDM *growth description has larger residual stress?*

The calculation is deliberately modest. It is a residual-structure diagnostic on a compact growth comparison set. It is not a Boltzmann-code calculation, not a joint cosmological fit, and not a claim of evidence for modified gravity.

The useful contribution is methodological: it tests whether the MAAT v1.2.1 diagnostic layer can be related to the structure of observed residuals in a reproducible way.

2 Data and Reference Model

The script uses a compact 13-point $f\sigma_8$ comparison set. The reference cosmology is

$$H_0 = 67.4, \qquad \Omega_{m0} = 0.315, \qquad \Omega_{\Lambda 0} = 0.685, \qquad \sigma_{8,0} = 0.811. \tag{1}$$

The reference expansion function is

$$E_{\Lambda\mathrm{CDM}}(z) = \sqrt{\Omega_{m0}(1+z)^3 + \Omega_{\Lambda 0}}, \tag{2}$$

and the approximate reference growth observable is

$$f_{\Lambda\mathrm{CDM}}\sigma_8(z) = \Omega_m(z)^{0.55} D(z)\sigma_{8,0}. \tag{3}$$

Here $D(z)$ is a Carroll–Press–Turner style analytic approximation, normalised to $D(0) = 1$.

This approximation is sufficient for a reproducible signature test. It should not be confused with a full perturbation-level likelihood analysis.

3 Residual Definitions

The raw residual is

$$r_{\mathrm{raw}}(z_i) = f_{\mathrm{obs}}\sigma_8(z_i) - f_{\Lambda\mathrm{CDM}}\sigma_8(z_i), \tag{4}$$

and the uncertainty-normalised residual is

$$r_\sigma(z_i) = \frac{f_{\mathrm{obs}}\sigma_8(z_i) - f_{\Lambda\mathrm{CDM}}\sigma_8(z_i)}{\sigma_i}. \tag{5}$$

The residual magnitude is

$$|r_\sigma(z_i)|. \tag{6}$$

Paper 40 compares structural diagnostics against both signed residuals and residual magnitudes. The signed residual test asks whether a diagnostic tracks direction. The magnitude test asks whether a diagnostic tracks stress intensity.

4 MAAT v1.2.1 Diagnostics

The projection template follows the same baseline construction as Papers 33, 34, and 39. It returns a normalised projection observable $\mathcal{C}_{\mathrm{proj}}(z)$ and a projection imbalance $R_{\mathrm{proj}}(z)$.

The v1.2.1 support fields are operational proxy supports:

- H: projection smoothness/coherence support,
- B: data-balance support from fractional $f\sigma_8$ residuals,
- S: projection activity support,
- V: connectedness support from bounded projection imbalance.

These support fields are not fundamental field measurements.

The v1.2.1 closure is

$$\boxed{R_{\mathrm{resp}} = (HBV)^{1/3}, \qquad R_{\mathrm{rob}} = \min\Big(R_{\mathrm{resp}}, (HBSV)^{1/4}\Big).} \tag{7}$$

The stability convention is

$$\mathrm{Stability} = R_{\mathrm{rob}}. \tag{8}$$

The two CCI diagnostics are

$$\mathrm{CCI}_{\mathrm{min}} = \frac{S}{H+B+V+\varepsilon}, \qquad \mathrm{CCI}_{\mathrm{diag}} = \frac{S}{H+B+V+R_{\mathrm{rob}}+\varepsilon}. \tag{9}$$

The implicit respect support R_{resp} is not included separately in the CCI denominator, avoiding double counting.

Important caveat. In this benchmark the balance support B is explicitly residual-sensitive. Therefore correlations involving $\mathrm{CCI}_{\mathrm{min}}$ or $\mathrm{CCI}_{\mathrm{diag}}$ are not fully blind predictions from an external theory. They are semi-supervised diagnostic consistency tests: if the structural layer is meaningful, the residual-sensitive support should organise residual stress in a coherent way rather than behave like a random field.

5 Correlation and Null Tests

For each diagnostic X, the script computes Pearson and Spearman correlations with

$$r_{\mathrm{raw}}, \qquad r_\sigma, \qquad |r_\sigma|. \tag{10}$$

Permutation p-values are computed with deterministic random seeds. For the main CCI magnitude test, the script also performs two robustness checks:

- a random-field null test, where the diagnostic ordering is permuted,
- a redshift-shuffle test, where the diagnostic is shuffled against the redshift-ordered target.

These tests do not turn the calculation into a precision likelihood. They simply check whether the observed diagnostic alignment is typical under simple permutation null models.

6 Results

6.1 Signed Residuals

The strongest signed-residual Spearman correlations occur for the projection imbalance channel and its connectedness counterpart:

$$\rho_{\mathrm{S}}(R_{\mathrm{proj}}, r_\sigma) = 0.6319, \qquad p_{\mathrm{perm}} = 0.0228, \tag{11}$$

and

$$\rho_{\mathrm{S}}(V, r_\sigma) = -0.6319, \qquad p_{\mathrm{perm}} = 0.0228. \tag{12}$$

This sign reversal is expected because V is constructed as a support associated with bounded projection imbalance.

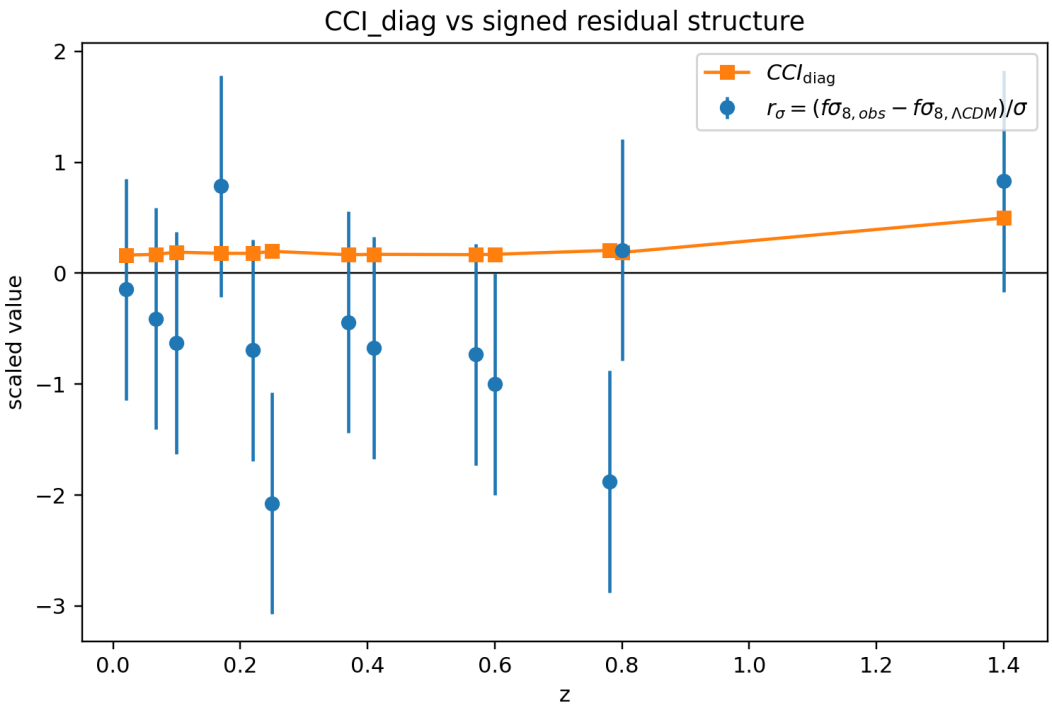

Figure 1: $\mathrm{CCI}_{\mathrm{diag}}$ and signed residual structure as a function of redshift. The diagnostic does not primarily track residual direction.

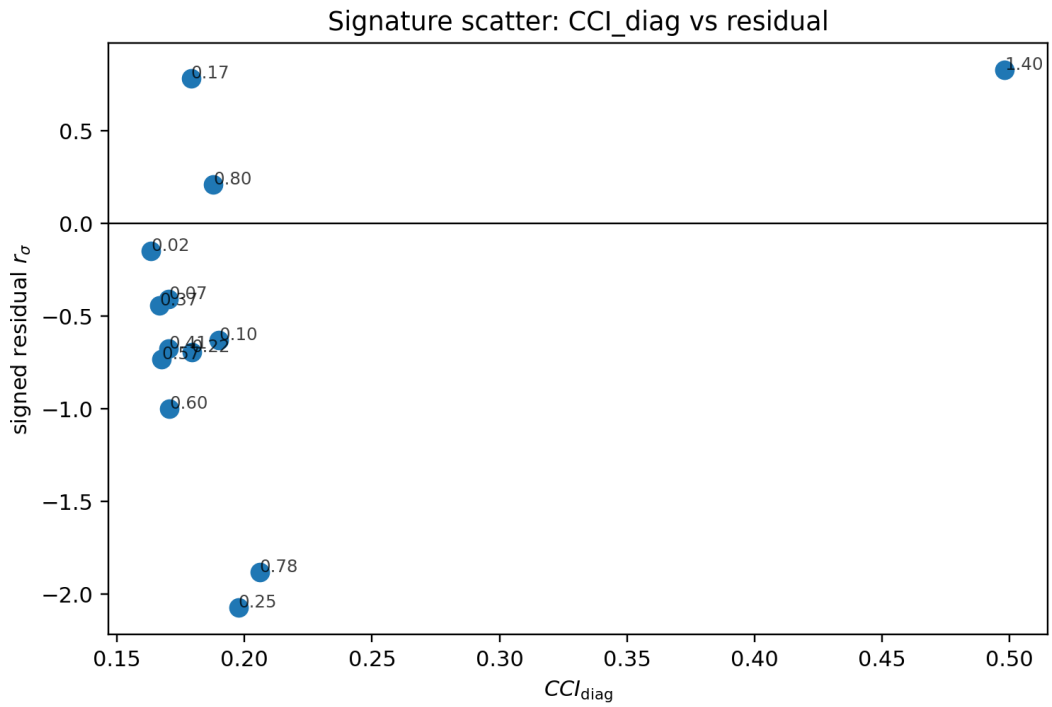

Figure 2: Scatter view of $\mathrm{CCI}_{\mathrm{diag}}$ versus signed normalised residuals.

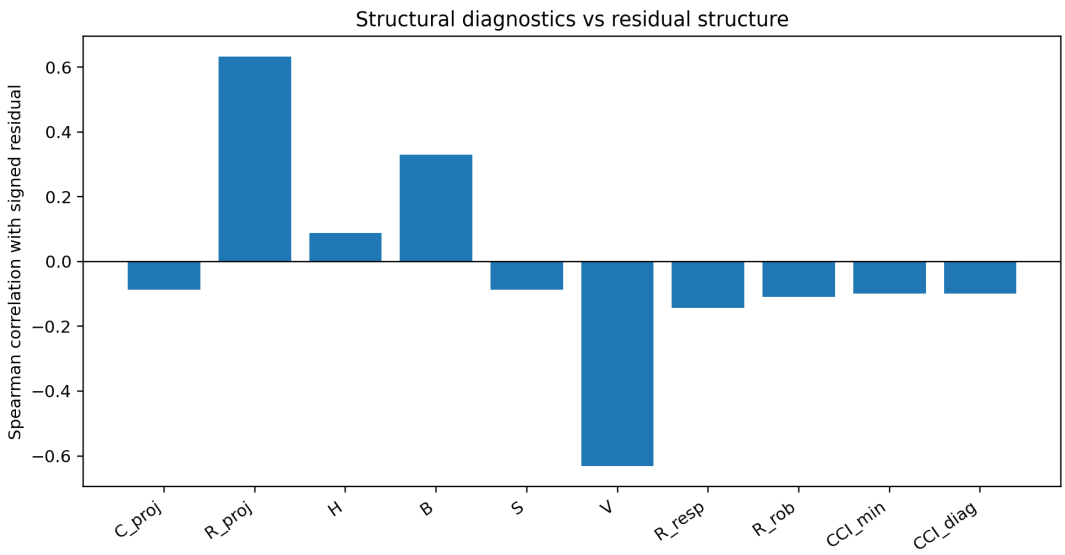

Figure 3: Spearman correlations between structural diagnostics and signed normalised residuals.

6.2 Residual Magnitudes

The CCI diagnostics show their clearest relation to residual magnitude:

$$\rho_{\mathrm{S}}(\mathrm{CCI}_{\mathrm{min}}, |r_\sigma|) = 0.5934, \qquad p_{\mathrm{perm}} = 0.0338, \tag{13}$$

and

$$\rho_{\mathrm{S}}(\mathrm{CCI}_{\mathrm{diag}}, |r_\sigma|) = 0.5934, \qquad p_{\mathrm{perm}} = 0.0338. \tag{14}$$

The balance support itself has a stronger anticorrelation with residual magnitude,

$$\rho_{\mathrm{S}}(B, |r_\sigma|) = -0.7912, \qquad p_{\mathrm{perm}} = 0.0017. \tag{15}$$

This is not surprising because B is constructed from residual information. It is nevertheless useful as an internal sanity check: higher residual magnitude lowers balance support, and the CCI responds coherently to that structural stress.

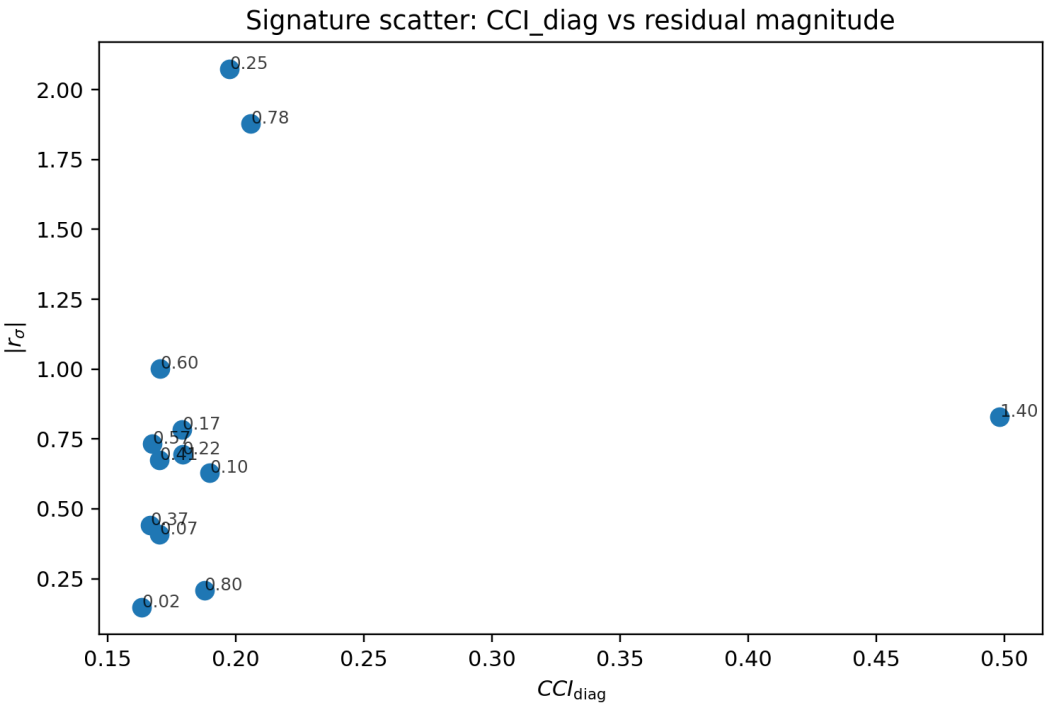

Figure 4: $\mathrm{CCI}_{\mathrm{diag}}$ versus residual magnitude. The observed monotonic alignment is exploratory because the sample size is small and the balance support is residual-sensitive.

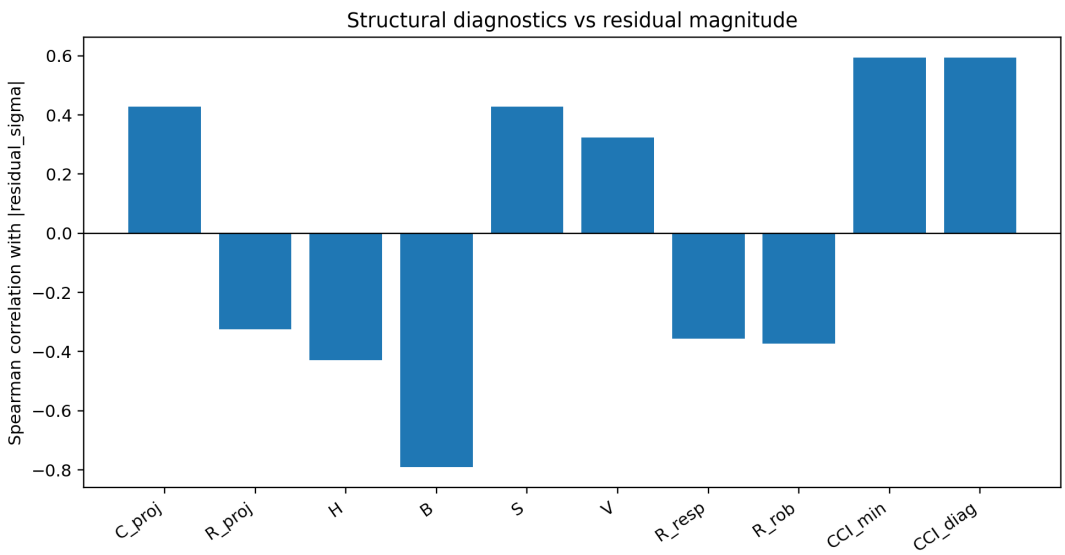

Figure 5: Spearman correlations between structural diagnostics and $|r_\sigma|$. The CCI diagnostics track residual magnitude more clearly than signed residual direction.

6.3 Null Tests

For the primary CCI diagnostic,

$$\rho_{\mathrm{S}}(\mathrm{CCI}_{\mathrm{diag}}, |r_\sigma|) = 0.5934. \tag{16}$$

The random-field null test gives

$$\langle \rho_{\mathrm{null}} \rangle = 0.0001, \qquad \sigma_{\mathrm{null}} = 0.2891, \qquad p_{\mathrm{null}} = 0.0363. \tag{17}$$

The redshift-shuffle test gives

$$\langle \rho_{\mathrm{shuffle}} \rangle = 0.0005, \qquad \sigma_{\mathrm{shuffle}} = 0.2891, \qquad p_{\mathrm{shuffle}} = 0.0359. \tag{18}$$

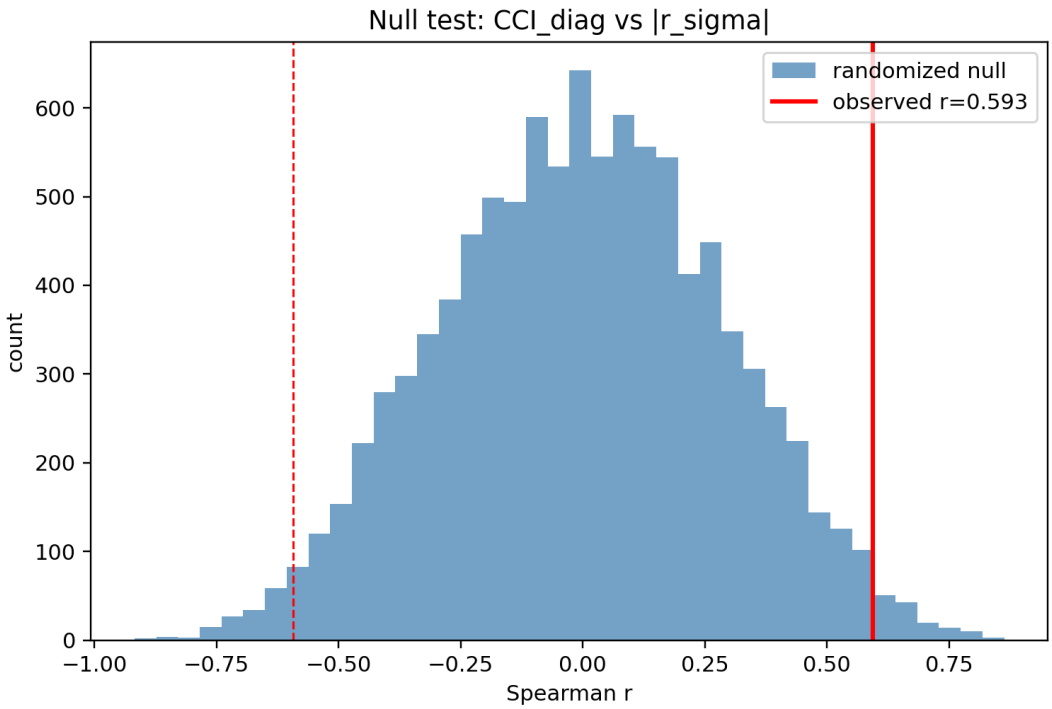

Figure 6: Permutation null distribution for the $\mathrm{CCI}_{\mathrm{diag}}$ residual-magnitude signature.

7 Interpretation

The result should be read as a structural diagnostic signal:

> *CCI does not predict the direction of growth residuals; it tracks their magnitude in this residual-sensitive benchmark.*

This supports a cautious interpretation of MAAT as a stress-diagnostic layer. It does not yet replace ΛCDM, improve a full likelihood, or generate a first-principles growth prediction. Instead, it identifies where a reference model exhibits larger residual stress and asks whether those regions are structurally organised.

The strongest claim justified by this paper is therefore:

> *The MAAT v1.2.1 diagnostic layer produces a reproducible, non-random-looking residual-magnitude signature on a compact growth-data comparison set.*

8 Limitations

- The sample size is small: $N = 13$ growth points.
- The $f\sigma_8$ baseline uses an analytic growth approximation, not a Boltzmann solver.
- No covariance matrix is used for the compact growth comparison set.
- The balance support B is constructed from residual information, so the CCI correlation is semi-supervised rather than fully independent.
- The reported p-values are exploratory permutation diagnostics, not survey-level significances.
- The result does not establish evidence for modified gravity, altered dark energy, or a detection of MAAT cosmology.

9 Reproducibility

The numerical pipeline is contained in the public repository:

https://github.com/Chris4081/structural-selection-principle

The experiment folder is:

`experiments/maat_paper40_structural_signature_test/`

Run:

```
python3 maat_paper40_structural_signature_test.py
```

The script generates:

- `outputs_paper40/paper40_summary.json`,
- `outputs_paper40/paper40_signature_table.csv`,
- `outputs_paper40/fig1_cci_diag_vs_signed_residual.png`,
- `outputs_paper40/fig2_scatter_cci_diag_signed_residual.png`,
- `outputs_paper40/fig3_scatter_cci_diag_abs_residual.png`,
- `outputs_paper40/fig4_spearman_signed_residuals.png`,
- `outputs_paper40/fig5_spearman_abs_residuals.png`,
- `outputs_paper40/fig6_breakthrough_null_test_cci_diag.png`.

10 Data Attribution and License Notes

The Planck-normalised reference parameters and compact $f\sigma_8$ comparison points are external scientific data and should be cited to the original publications/collaborations when reused or discussed. The CSV tables and figures in this repository are derived analysis artifacts generated for reproducibility of the MAAT diagnostic calculation. No endorsement by the Planck Collaboration, survey collaborations, or original data authors is implied.

The repository code is released under the repository license.

11 Conclusion

Paper 40 moves the MAAT cosmology sequence from observable modulation toward residual-structure diagnostics. On a compact $f\sigma_8$ comparison set, the diagnostic CCI correlates with the magnitude of ΛCDM residuals, and this alignment is not reproduced by simple random-field or redshift-shuffle null tests at the exploratory level.

The result is not a detection. It is a reproducible structural-signature benchmark that suggests a concrete next step: replace the residual-sensitive proxy construction by a fully independent perturbation-level prediction and test it with survey covariance matrices.

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